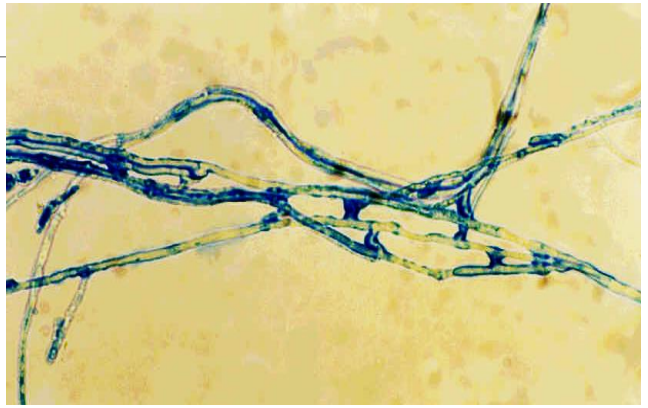
FUNGAL INFECTIONS

By

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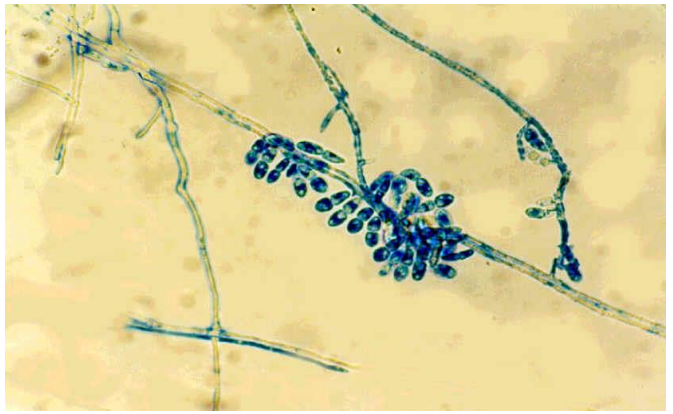
Causative organisms



1- Dermatophytes

- **Multicellular organisms form long filaments called hyphae.**

It includes 3 genera: Microsporum, Trichophyton & Epidermophyton.



2-Yeasts



- **Unicellular like organisms.**
- **Like: - Candida → candidiasis**

  - Malassezia → pityriasis versicolor

Dermatophyte infection

Clinical Types of dermatophyte Infection

According to affected site:

- **Tinea capitis → scalp.**
- **Tinea corporis (circinata) → body.**
- **Tinea cruris → groin.**
- **Tinea pedis → feet.**
- **Onychomycosis→ Nails**

Tinea Capitis

- **Mainly in school children, more in boys**

- **Modes of infection:**
  - **Direct contact with infected child or animal.**
  - **Indirect (the use of patient's fomites, brushes, caps).**

Clinical Types Of Tinea Capitis

1- Scaly type (scaly ringworm).

2- Kerion.

3- Favus.

1. Scaly type (scaly ringworm)



- **Lesion:**

  Single or multiple, round or oval, scaly (grayish white) patches.

- **Hair:**

  broken off (2-3mm long), dull-grey (lusterless)



- **Resolution: without scar**

2- Kerion



- **Lesion:** **Inflammatory, Boggy indurated swelling, with crusting, resembles pyogenic abscess.**

- **Hairs overlying it:** **loose, on removal of hair → seropus.**

- **Resolution:** **may lead to scar (cicatricial alopecia).**

- **In extensive lesions :**

  Fever, pain & regional lymphadenopathy may be present.



Diagnosis



1- Clinical Picture

2- Wood's light:

Certain fungi fluoresce when examined by Wood's light:

3- Direct microscopic examination: of infected hairs and scales --- hyphae and spores.

4- Culture.

Treatment of tinea capitis

A- Topical treatment:

- **Whitfield's ointment.**
- **Imidazoles**, e.g.clotrimazole, econazole, ketoconazole.
- **Allylamines** e.g. terbinafine

B- Systemic treatment:

- **Griseofulvin (**10 mg/kg/d. for 6- 8 weeks. in favus)
- **Itraconazole**
- **Fluconazole**
- **Terbinafine**

Tinea Corporis (circinata)

➢**Age:** more common in children

➢**Lesion:** single or multiple, circular lesions, annular lesions, erythematous, raised edge, more inflamed than center "active edge"

➢**Treatment:**

- Mild: Topical antifungal 2-4 weeks.
- Extensive: Systemic antifungals





Tinea Cruris



- **It is a dermatophyte infection of the groin.**

- **Lesion: erythematous scaly patch → well-defined active edge.**
- **Symptom: Itching is common.**
- **Sex: More common in men.**
- **Weather: hot summer.**



- **Treatment:**

- **The area should be kept dry**
- **Wearing loose underclothing**
- **Topical antifungal powder**
- **Systemic antifungals (2-4 weeks).**

Tinea Pedis (Athlete's foot)

- **It is ringworm of the feet.**
- **Most common fungal disease.**
- **Predisposing factors: Wearing sockets & shoes for long duration, hyperhidrosis, communal showers, swimming baths.**
- **Age &sex: common in adult male.**

➢ It has 3 clinical varieties:

Interdigital type:

- **Commonest.**
- **Clinical picture: peeling, maceration & fissuring.**
- **Site: lateral toe clefts.**



➢ Treatment:

- **Dry the feet thoroughly after bathing.**
- **Antifungal powder for susceptible persons.**
- **Topical antifungals for mild cases.**
- **Systemic antifungals for extensive and chronic cases.**

Yeast infection

Pityriasis versicolor
(Tinea versicolor)



- **Caustive organism:** yeast **Malassezia** (normal flora of the skin).
- **Climate**: tropical climate, warmer months
- **Age:** young adults.
- **Site:** chest, neck, upper arm, abdomen, back
- **Lesion:** macules covered by fine branny scales, or coalesce into large confluent patches.
- **Color:** yellowish or brownish (in white skin), hypopigmented (in dark skin).



➢ Treatment:

1. Topical antifungals daily for 2-3ws.

2. Ketoconazole 2% shampoo.

3. Selenium sulphide 2-5% shampoo

4. Sodium hyposulphite 20% solution

5. Systemic antifungals: fluconazole, itraconazole (*selected cases*).

NB. Oral terbinafine and Griseofulvin are ineffective

Candidiasis

- **Causative organism:** **Candida albicans (Yeast) (commonest).**
- **Normal inhabitant: at various sites (gut, mouth and vagina).**
- **Pathogen: in the presence of impaired immune response, or in body folds.**
- **Predisposing factors:**
  - **Warmth, moisture, maceration of the skin**
  - **Drugs: prolonged use of systemic steroids, antibiotics, immunosuppressives.**
  - **Chronic debilitating diseases :e.g. diabetes mellitus, lymphoma, AIDS.**

Clinical variants:



1- **Oral candidiasis** (thrush)

- Creamy, white patches (pseudomembrane).
- On removal → erythematous base.

2- **Candidal vulvovaginitis**

Erythema, maceration, pruritis, genital discharge

3- **Candidal intertrigo**

- Skin folds → obese.
- Itchy, moist, erythematous patches, with satellites outside edge.



4- **Napkin candidiasis** (Diaper rash)

- It is enhanced by maceration from wet diapers.



Treatment:

- **Topical nystatin**
- **Topical imidazole**
- **Systemic: Fluconazole, Itraconazole**


QUIZ

True or false

1- Malassezia is the causative organism of tinea pedis.

2- Tinea capitis is caused by dermatophytes.

3- Tinea cruris is more common in females.

4- Onychomycosis is only caused by Candida.

Case 1

A six years old boy presented with two rounded bald scaly patches in the scalp. The hair destructed at the scalp surface with no scar formation. The diagnosis is:-

a- Scaly ringworm

b- Alopecia areata

c- Favus

Case 2

A 10 years old boy presented with painless boggy swelling on the scalp. Hair overlying it is loose. On Removal of hair, seropus discharge appear. Lymph-adenopathy is absent. The case may be:

a- Favus.

b- Kerion.

c- Abscess.


Thank you!